

AI-Driven adaptive Coding Assistance and Interview Preparation Platform

Project ID : 25-26J-364

Project Proposal Report

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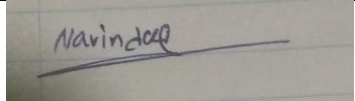
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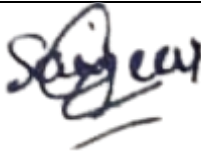
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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Ms. Sanjeevi Chandrasiri	
Date	2025/08/31

ABSTRACT

Interview preparation is a significant challenge for university students, especially in fields like software engineering. In these areas, strong technical skills must go hand in hand with good communication and interpersonal abilities. Traditional methods, such as instructor-led mock interviews and peer training, require many resources and do not easily scale. Current AI tools, like Google's Interview Warmup and commercial options such as HireVue, offer some solutions but do not provide complete, personalized, and immediate feedback.

This research suggests creating a Real-Time AI Voice Agent Interview Platform aimed at university students getting ready for technical and behavioral interviews. The system consists of three main parts: (1) AI-generated questions that adjust from basic to advanced levels based on student performance; (2) speech-based interaction using automatic speech recognition (ASR) and text-to-speech (TTS) technologies to create a realistic interview setting; and (3) facial emotion detection using convolutional neural networks (CNNs) to pick up non-verbal signals like confidence, nervousness, or neutrality.

The platform allows for an interactive experience where students answer questions verbally while the AI agent assesses both their spoken and non-verbal communication. After each session, the system provides structured feedback, including an interview score, a detailed performance analysis, and personalized suggestions for improvement. Unlike traditional systems, this proposed solution focuses on a well-rounded evaluation by considering technical accuracy, communication clarity, and emotional composure at the same time.

The expected outcome of this research is a scalable and accessible interview preparation tool that improves student readiness for job interviews. By connecting the assessment of technical knowledge with feedback on non-verbal communication, this platform aims to offer a fuller preparation experience, ultimately boosting student employability and confidence.

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LIST OF ABBREVIATIONS

Abbreviation	Description
CNN	Convolutional Neural Network
API	Application Programming Interface
STT	Speech-to-Text
FER	Facial Emotion Recognition
TTS	text-to-speech

1. INTRODUCTION

In today's job market, to get a job, you need more than just a degree: you need to have knowledge, skills in communicating, problem-solving, and professional behavior. Software engineering students and students from related, technical disciplines at university struggle preparing for professional interviews. Although conventional preparation techniques like mock interviews, written assignments, or online resources can help covering the fundamentals – they don't necessarily offer the personal touch, immediate feedback, or the inclusion of the non-verbal communication aspect in the practice. **Real-Time AI Voice Agent Interview Platform** The Real-Time AI Voice Agent Interview Platform addresses this void in technology and serves as an innovative, interactive, and dynamic means of interview preparation.

Featuring innovative technologies such as: **artificial intelligence** (AI), voice processing, and facial emotion recognition, the platform mimics the same environment one would experience during a physical and in-person interview which is so crucial in the rental industry. Through allowing students to conduct natural voice-based communication with an AI agent and record their facial expressions with a webcam, a better immersive condition is reached. As opposed to existing techniques, which predominantly test technical knowledge, this system tests a student's verbal articulating, his/her emotional state, his/her confidence and his/her problem-solving abilities simultaneously.

This combination helps provide a more comprehensive assessment of interview preparedness. At the heart, I see the AI Interview Agent, based on big language ones, e.g., Gemini or ChatGPT. The tutor generates interview questions along with their order and complexity descriptively on-the-fly, adjusts question difficulty depending on the student's past performance, and maintains dialog coherence with follow-up questions and after-ask clarifications. Additionally, the Facial Emotion Detection Module leverages deep learning methods (such as CNNs) to detect emotions such as happiness, anxiety, neutrality, or stress. This non-verbal aspect of their performance helps assess how students present themselves to under pressure, and provides a type of feedback that other interview preparation tools often miss out on. The platform also integrates the speech-to-text (STT) and text-to-speech (TTS) to

facilitate voice-based close-loop interaction. Students answer AI-generated questions orally, while the AI agent asks questions using natural speech. This live voice interaction not only makes the interview simulation more realistic, but students have an opportunity to practice being clear, the tone, rate, and over-all effectiveness/style of communication

1.1 Background & Literature survey

The transition from university to professional life is arguably the most daunting task facing undergraduates, particularly software engineering. Employers expect graduates not only to possess great technical skills in areas such as algorithms, programming, databases, and software design but to have great soft skills as well. These include communication skills, the ability to perform well under pressure, and emotional resilience. Interviews represent the fundamental gateway between undergraduate students and professional careers, and performance in these pressure-cooker situations often determines employability.

The majority of undergraduates, especially those conducting interviews for their first internships or entry positions, struggle to perform competently in interviews even if they have superior academic qualifications. Common issues are nervousness, inconsistent performances, and scarcity of experience in formal interview settings. Students generally have trouble managing stress, interpreting interviewer cues, or structuring their answers effectively. Traditional preparation techniques such as mock interviews led by classmates, workshop sessions organized by career services, or banks of questions don't provide sufficient exposure and fail to generate dynamic realism associated with a real interview. They don't even provide continuous, individualized feedback, which is crucial for steady improvement.

Artificial Intelligence (AI) development has opened up new possibilities to overcome these issues. AI-based systems such as intelligent tutoring systems, conversational agents, and adaptive learning environments have shown immense potential in education and training. These systems can model actual problems, provide immediate tailored feedback, and accommodate repeated

practice without the necessity of having human mentors. Regarding interview preparation, AI-based platforms have been adopted that simulate employer line of questioning, grade verbal responses, and offer patterned tests.

But most interview preparation programs today only emphasize technical correctness, linguistic ability, or the ability to provide the "correct" answer. They ignore the equally vital component of non-verbal communication and emotional quotient. But employers again and again validate that body language, eye contact, confidence, and demeanor are the deciding factors in candidate performance. Students who cannot effectively regulate their emotions in an interview perform less effectively compared to their real potential.

Facial Emotion Recognition (FER) using computer vision and deep learning presents a key to solving this issue. Detecting smiles of happiness, nervous glances, stress, or neutrality, FER systems can detect key non-verbal signals. Integrating such analysis into interview training would enable students to receive beneficial feedback on both what they are saying and how they are appearing. Combined with adaptive questioning styles that progressively increase challenge in accordance with the candidate's performance, this collaboration could constitute an interview simulation environment that mirrors real-world dynamics.

Hence, the development of an AI-powered, real-time interview platform with voice interaction, adaptive questioning, and facial emotion recognition has a lot of promise to prepare software engineering undergraduates. The system can enable students to have real-like practice, reduce anxiety, and enhance technical and emotional readiness for competitive recruitment.

Literature Survey

Application of Artificial Intelligence (AI) in education, training, and professional development has grown immensely in recent years. One significant application is AI-driven mock interviewing systems, which simulate employer-style questioning through voice or text interfaces and grade candidates on correctness, fluency, and coherence [1]. For example, such services as Google Interview Warmup provide question banks and automated comments to help students practice [2]. While such systems assist in training students with structured interviews, they tend to be non-adaptive to learning in terms of individual progress and are technical or linguistic in focus with minimal regard for behavioral and emotional inputs.

Facial Emotion Recognition (FER) has evolved human-computer interaction and affective computing. Employing convolutional neural networks (CNNs) and deep learning, FER systems are capable of analyzing emotions like happiness, sadness, fear, and nervousness from live video feeds [3]. Libraries like OpenCV and face-api.js have enabled the deployment of lightweight real-time FER applications in education, healthcare, and online testing. Application of FER with mock interview platforms is limited, though, while opportunities exist to provide feedback on confidence and emotional stability [3].

Adaptive learning system studies give more information. Adaptive systems continuously adapt task difficulty based on learner performance, with incremental skill acquisition [4]. In interview scenarios, adaptive questioning can mirror real employer practice, ranging from simple to complex technical and behavioral questions, with increased retention of knowledge, motivation, and confidence.

Automated feedback and scoring capabilities bring further sophistication. AI systems can analyze free response and offer helpful feedback, though the majority treat only technical or linguistic correctness and not multimodal signals such as tone, delivery, and facial expressions [5].

Overall, recent research attests to progress in AI interviews, FER, adaptive questioning, and automated feedback. They are all, however, specialized systems focused on one component and

not integrated into one single platform. This implies that what is needed is a Real-Time AI Voice Agent Interview Platform that combines adaptive questioning, facial emotion recognition, and automated feedback in order to better prepare students for professional interviews [1]–[5].

1.2 Research Gap

Even though there have been significant advances in AI-powered interviewing platforms, face emotion recognition (FER), adaptive learning, and machine-based feedback, current solutions remain narrow and fragmented. Platforms such as Google Interview Warmup focus primarily on assessing verbal accuracy and language skills but fail to incorporate non-verbal aspects such as confidence, facial expressions, and stress [1], [2]. FER systems have demonstrated the ability to recognize emotions in real time, but they are rarely integrated into simulated interview training systems [3]. Adaptive learning methods can adjust the difficulty level of questions, yet most simulated interview platforms remain static, treating all users similarly [4]. Automated feedback is available, but it is typically technical or performance-based and lacks multimodal insight such as tone of voice or emotional state [5].

To illustrate, current systems face several limitations:

1. **Modality fragmentation.** Text-only mock interviews lack real conversational dynamics; video analytics platforms often require uploads and provide delayed rather than interactive feedback.
2. **Weak non-verbal integration.** Many tools ignore facial expressions, eye contact, or stress indicators, or treat them as afterthoughts; FER prototypes often run offline and are not aligned with live dialogue.
3. **Limited personalization.** Few systems tailor questions to a student’s background (courses, projects) or adapt difficulty in-session based on both performance and affect.
4. **Shallow feedback.** Feedback often focuses on keywords or generic recommendations; alignment between what was said (content), how it was said (prosody), and how it appeared (facial affect) is rare.

In summary, there is a well-recognized research gap in developing an end-to-end, real-time interview practice system that:

- Uses adaptive questioning to progressively challenge students.
- Employs facial emotion recognition to provide feedback on emotional state and confidence.
- Delivers multimodal automated feedback that integrates verbal performance, tone, and affect.

Bridging this gap would enable university students—particularly software engineering interns—to experience more realistic, personalized, and emotionally intelligent interview practice, ultimately improving their readiness for real-world professional interviews.

Feature / Platform	Google Interview Warmup	HireVue	Comments / Observations
Real-Time Voice Interaction	✓ Yes	✓ Yes	Both platforms provide voice-based Q&A, allowing candidates to practice verbal communication; however, the interaction is primarily one-way and lacks advanced dialogue management.
Facial Emotion Recognition (FER)	✗ No	✓ Limited	FER is either absent or limited, meaning the platforms cannot provide feedback on candidate emotions or confidence during responses, which is a critical non-verbal component of interviews

Adaptive Questioning	✓ Basic	✗ No	Google Warmup provides minimal question adaptation based on past answers, while HireVue does not adjust question difficulty dynamically.
Personalized Difficulty Levels	✗ No	✗ Limited	Both platforms fail to fully personalize difficulty levels based on candidate skill, profile, or performance, limiting tailored preparation.
Confidence Assessment	✗ No	✗ Limited	Assessment of candidate confidence or delivery style is absent or very basic, preventing comprehensive feedback on soft skills.

Table 1. Comparative Feature Analysis of Existing Platforms

1.3 Research Problem

Preparing for professional interviews is a critical yet challenging aspect of a university student's transition from academics to the workforce, particularly for software engineering interns. While conventional mock interview platforms offer opportunities to practice, they primarily focus on assessing verbal responses, technical knowledge, and language proficiency. However, interviews in real-world professional settings involve a complex interplay of verbal and non-verbal cues, including confidence, facial expressions, tone of voice, and emotional control [1], [2]. Current platforms fail to capture these nuances, leaving students underprepared for the interpersonal and behavioral aspects of interviews.

Existing AI-powered mock interview systems, such as **Google Interview Warmup** [3] and other automated platforms [1], provide basic automated scoring and feedback but largely overlook the emotional intelligence dimension. Facial Emotion Recognition (FER) systems exist independently [4], and adaptive learning techniques can adjust question difficulty [5]. However, very few platforms integrate these elements in a seamless, real-time environment. Moreover, the feedback provided by most platforms is either technical or performance-based, lacking multimodal insights that could guide students on how to modulate tone, maintain composure under stress, or convey confidence effectively [2], [6], [7].

Recent research highlights the value of multimodal learning analytics and emotion-aware educational technologies [8], [9]. Similarly, AI in recruitment and assessment is gaining global adoption, but ethical and interpretability issues remain underexplored [10]. This creates a significant gap in developing a comprehensive, real-time, emotionally intelligent interview preparation system that simultaneously addresses verbal skills, non-verbal cues, adaptive learning, and multimodal feedback. Without such integration, students are unable to experience realistic simulations of professional interviews, limiting their ability to identify personal weaknesses, build confidence, and refine their overall performance.

2. OBJECTIVES

2.1. Main Objective

The main objective of this research is to **design, develop, and evaluate a comprehensive, real-time AI-powered interview platform** that holistically addresses the challenges faced by university students, particularly software engineering interns, in preparing for professional interviews. The platform aims to integrate **advanced artificial intelligence techniques**, including speech recognition, natural language processing, facial emotion recognition, and adaptive learning, to deliver a fully immersive and personalized mock interview experience.

By combining **verbal and non-verbal evaluation mechanisms**, the system seeks to go beyond conventional assessment approaches that focus solely on language skills or technical accuracy. It will analyze students' responses in real-time, assess their **emotional state, confidence, and stress levels**, and provide **multimodal feedback** that is both actionable and tailored to the individual's performance. The platform intends to progressively challenge students through **adaptive questioning**, dynamically adjusting the difficulty of questions based on prior responses and performance trends, ensuring that students are continuously engaged and improving

Furthermore, the proposed system will simulate real-world interview scenarios, providing a safe yet realistic environment where students can practice and enhance their soft skills, technical knowledge, and behavioral responses. The primary objective is to improve students' overall preparedness, strengthening not only their ability to answer questions effectively but also their confidence, emotional resilience, and professional presentation

In essence, this research seeks to bridge the gap between existing AI mock interview platforms and the comprehensive needs of modern students, offering an emotionally intelligent, adaptive, and holistic solution that better prepares them for real-world job interviews. In doing so, the platform aims to drive educational innovation, enhance student employability, and establish a benchmark for future advancements in AI-driven interview preparation systems

2.2. Sub-Objectives

The specific objectives of this research are designed to **operationalize the main objective** by defining measurable and achievable goals that guide the development and evaluation of the AI-powered interview platform. These objectives focus on **verbal interaction, facial emotion detection, adaptive questioning, and multimodal feedback** to create a comprehensive interview preparation system

1. Develop a real-time voice-based interaction system
 - Implement **Text-to-Speech (TTS)** for AI-generated questions and **Speech-to-Text (STT)** for capturing user responses.
 - Ensure seamless **bidirectional communication**, creating a realistic mock interview environment.
2. **Integrate facial emotion recognition (FER)**
 - Use FER to detect **user emotions such as happiness, neutrality, or stress** during responses.
 - Provide real-time analysis of **confidence, engagement, and emotional state**, forming the basis for adaptive questioning and feedback.
3. **Implement adaptive questioning mechanisms**
 - Dynamically adjust the **difficulty, type, and sequence of interview questions** based on previous answers and detected emotions.
 - Ensure personalized progression from **basic to advanced questions**, maintaining an appropriate challenge level for each user.
4. **Provide multimodal automated feedback**
 - Deliver **voice-based feedback** that combines evaluation of verbal responses with insights from facial emotions.
 - Highlight areas for improvement, including **answer structure, clarity, confidence, and emotional regulation**

3. METHODOLOGY

The methodology of the Real-Time AI Voice Agent Interview Platform is designed to create a simulated interview experience for university students, particularly those in software engineering. The system combines **voice communication, face emotion detection, personalized question generation, and automated scoring** to provide realistic and useful feedback. This section explains each component and how they interact to form a complete system.

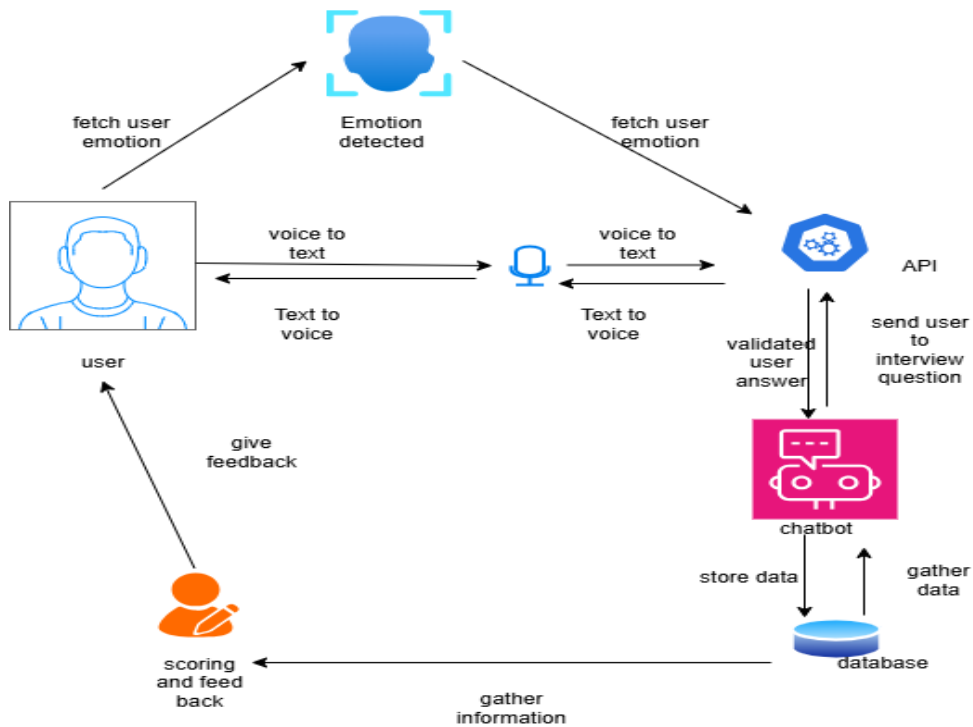


Figure 1.system diagram

User Interaction Layer

The process begins with the student who interacts directly with the AI platform. Students communicate through **voice**, making the interview simulation feel natural and close to a real scenario. Instead of typing answers, users respond verbally, just as they would in a professional interview. The system also uses a webcam to capture facial expressions during the conversation. This dual input – voice and facial emotions – helps create a richer and more authentic interaction experience.

Voice Processing: Speech-to-Text (STT) and Text-to-Speech (TTS)

Since the user communicates by voice, the system first converts spoken words into text using **Speech-to-Text (STT)** technology. This ensures that the AI agent can understand the content of the student's responses accurately. On the other side, when the AI asks a question, the system uses **Text-to-Speech (TTS)** to generate spoken output. This two-way voice processing enables smooth real-time communication between the student and the AI agent. Tools such as VAPI (Voice API) can be integrated to handle both STT and TTS functions effectively.

Natural Language Processing (AI Interview Agent)

At the core of the platform is the **AI interview agent**, powered by a Large Language Model such as Google Gemini or ChatGPT. The AI generates questions that range from simple to advanced, based on the student's responses and performance. It evaluates the content of the student's answers for accuracy, completeness, and relevance. This adaptive question generation helps simulate the progressive difficulty of a real interview. The AI also considers detected emotions when framing follow-up questions—for example, offering encouragement if the student looks stressed or asking for clarification if the student seems confused.

Emotion Detection (Facial Analysis)

In addition to analyzing spoken answers, the system uses **computer vision techniques**, such as CNN-based face recognition or face-api.js, to interpret facial expressions. Emotions like happiness, neutrality, or discomfort are identified in real time and sent to the AI interview agent. These cues allow the system to better understand the student's confidence and communication style. For instance, if a student consistently looks nervous, the feedback may include advice on maintaining eye contact or managing stress during interviews.

Data Processing and Storage

All data—including transcribed answers, detected emotions, AI evaluations, and timestamps—are stored in a secure database. This allows for detailed record-keeping and supports further analysis. The stored data can later be used for training purposes, progress tracking, or improving the AI model itself. By structuring the data carefully, the system ensures that every interview session is both reproducible and measurable.

Scoring and Feedback Generation

The final stage of the methodology involves **evaluation and scoring**. The system generates two main types of scores:

- **Verbal Performance Score** – Based on clarity, accuracy, and completeness of spoken answers.
- **Non-Verbal Performance Score** – Based on emotional stability, confidence, and engagement observed through facial expressions.

These two dimensions are combined into an **overall interview score**, which gives the student a clear measure of performance. Along with the score, the AI generates **personalized feedback**, highlighting strengths (e.g., strong technical knowledge) and weaknesses (e.g., nervous body language). This ensures the student knows exactly where to improve.

Final Report Delivery

At the end of the session, the platform generates a structured report for the student. This report includes:

- Interview questions asked and the student's responses.
- Detected emotions at key points during the interview.
- Strengths and areas for improvement.
- Final performance score.

This report serves as a learning tool, enabling students to reflect on their practice session and prepare better for real-world interviews.

Work-break Down Structure

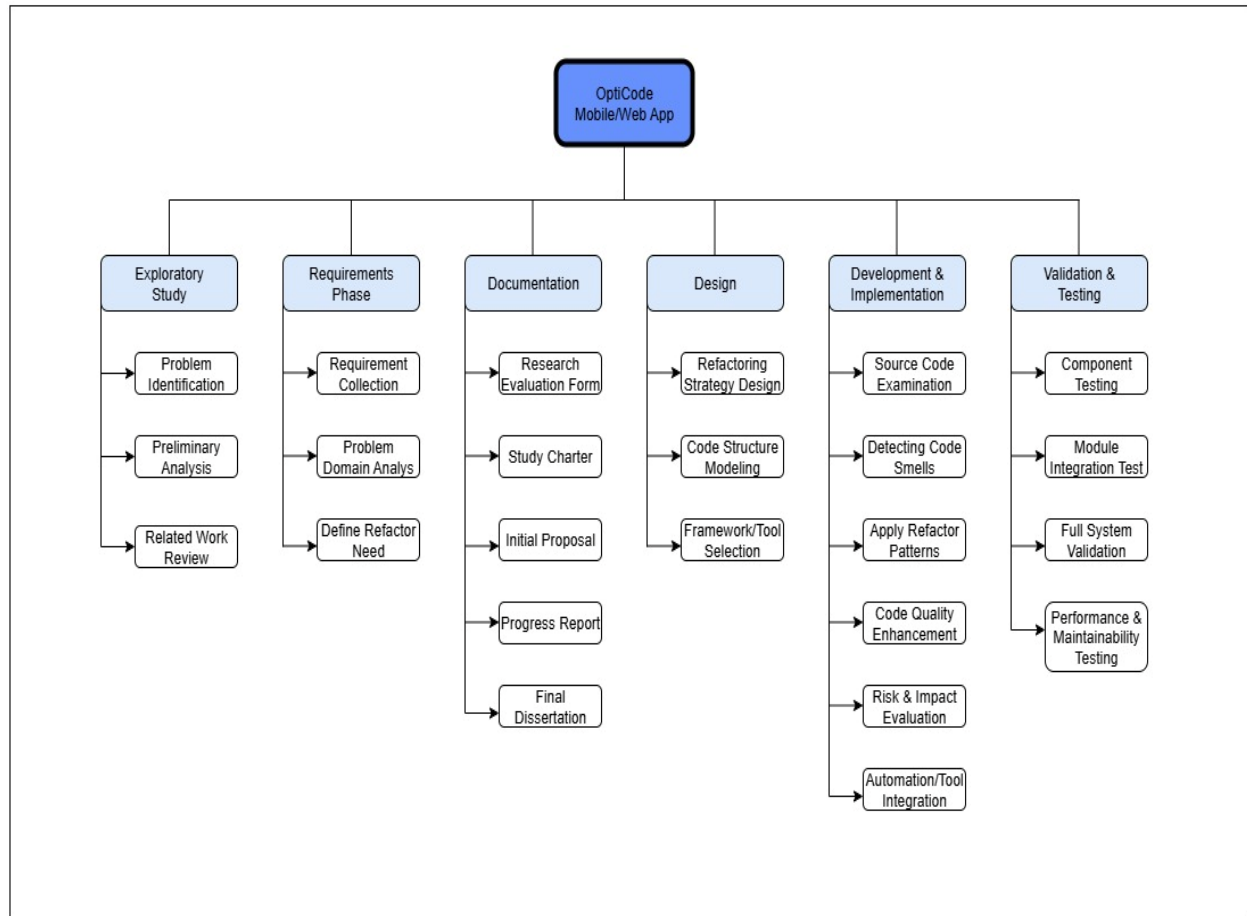


Figure 2.: Work Breakdown

4. PROJECT REQUIREMENTS

4.1 Functional requirements

.User Authentication and Profile Management

- Students must be able to create accounts using email, Google OAuth2, or university credentials.
- The system should allow secure login and logout, with role-based access for students, administrators, and instructors.
- Users can update personal information, preferences, and learning history.

Voice-Based Interview Interaction

- Students interact with the system using voice input rather than text.
- The system converts spoken responses into text using **Speech-to-Text (STT)** technology for processing.
- AI responses are converted into natural-sounding voice output using **Text-to-Speech (TTS)**.

AI-Powered Interview Question Generation

- The system dynamically generates interview questions based on the student's skill level, performance history, and progression.
- Question types range from simple knowledge-based questions to advanced scenario-based or problem-solving tasks.
- The AI provides follow-up questions and prompts for clarification, ensuring natural interview flow.

4.Facial Emotion Detection

- The system captures facial expressions using a webcam.
- Deep learning models such as **Convolutional Neural Networks (CNNs)** detect emotions like confidence, nervousness, happiness, neutrality, or stress.

- Emotion data is integrated into the scoring engine to provide non-verbal insights during evaluation

4.2 Non-functional requirements

Non-functional requirements define **how the system performs** rather than what it does. They are critical to ensure a smooth, reliable, and user-friendly experience. For your AI interview platform, the key non-functional requirements include:

Usability

- **Definition:** Usability measures how easy and intuitive the platform is for students to interact with.
- **Importance:** University students, especially those with limited technical expertise, should be able to navigate the platform effortlessly, access mock interviews, and understand feedback.
- **Examples in your system:**
 - Simple and intuitive web interface.
 - Clear voice prompts and easy-to-follow instructions.
 - Quick access to past interviews and feedback reports.

Performance

- **Definition:** Performance refers to how efficiently the system executes tasks, including response time, processing speed, and handling concurrent users.
- **Importance:** AI-driven voice interactions, facial emotion recognition, and adaptive question generation require low-latency processing for a realistic interview experience.
- **Examples in your system:**
 - Minimal delay in AI voice responses.
 - Real-time analysis of facial expressions and voice tone.
 - Fast retrieval of adaptive questions based on user performance.

Reliability

- **Definition:** Reliability is the system's ability to function correctly and consistently over time without failures.

- **Importance:** Students depend on the platform for interview preparation, so interruptions or errors could disrupt learning and reduce confidence.
- **Examples in your system:**
 - Consistent performance during multiple sessions.
 - Accurate processing of voice input and facial emotion data.
 - Error handling mechanisms in case of network or hardware failures.

Scalability

- **Definition:** Scalability refers to the platform's ability to handle increasing numbers of users or workload without performance degradation.
- **Importance:** As more students use the system simultaneously, especially during peak periods like exam preparation or internship seasons, the system should remain responsive.
- **Examples in your system:**
 - Ability to support hundreds of concurrent student interviews.
 - Dynamic resource allocation for AI processing tasks.
 - Cloud-based deployment to scale processing for multiple sessions.

Availability

- **Definition:** Availability measures the system's uptime or readiness for use. High availability ensures that students can access the platform whenever needed.
- **Importance:** A platform that is frequently down or inaccessible undermines trust and usefulness.
- **Examples in your system:**
 - 24/7 accessibility for remote students.
 - Backup servers or failover mechanisms to minimize downtime.
 - Monitoring tools to detect and resolve issues promptly.

5. Overall Diagram

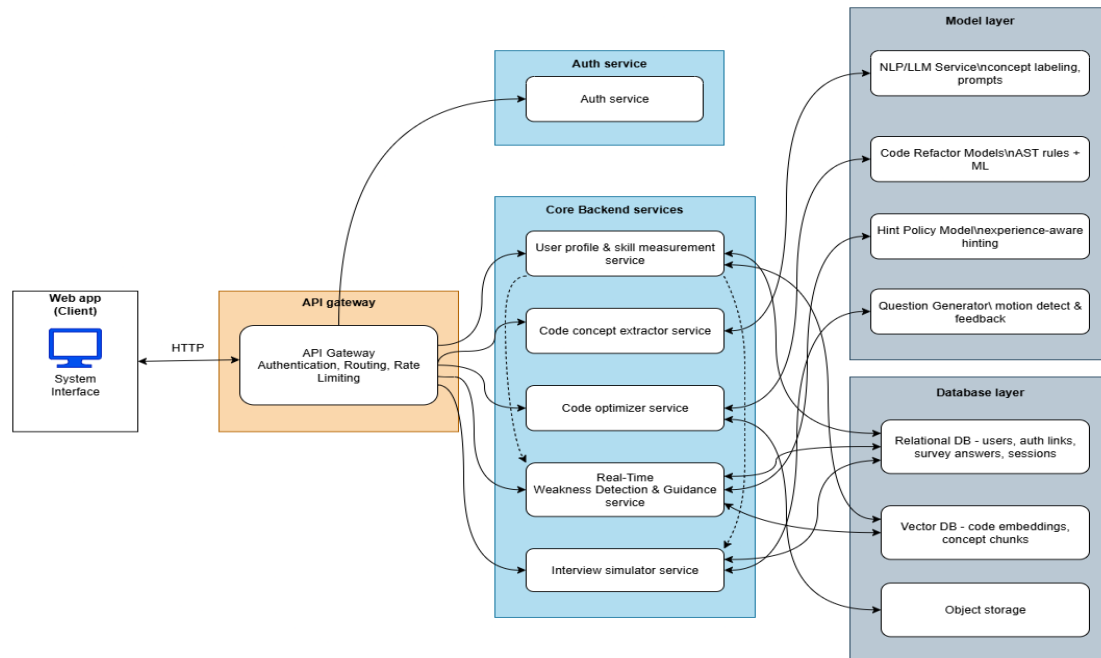


Figure 3. Overall System Diagram

Gantt chart

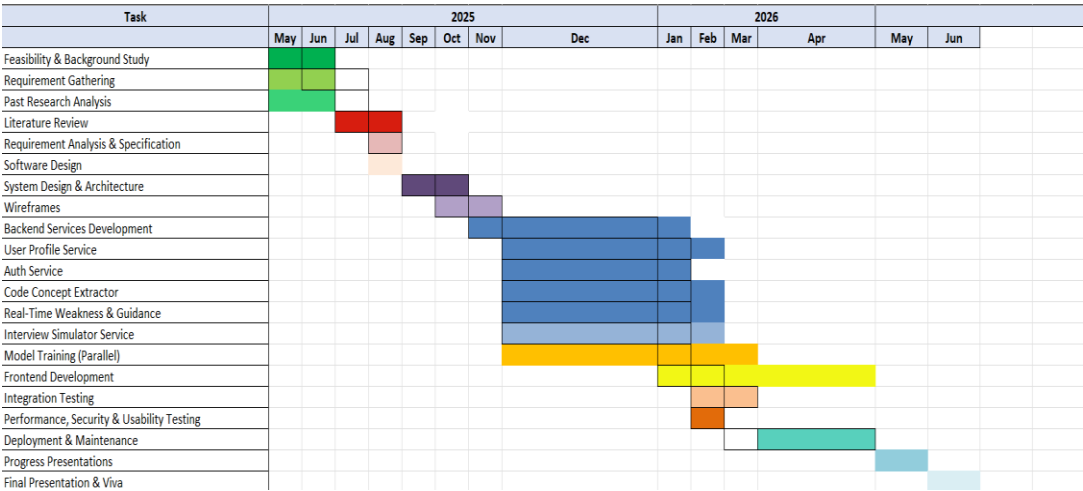


Figure 4. Gantt chart

REFERENCE

- [1] R. Patel, “AI-driven mock interview platforms: A review,” *Int. J. Artif. Intell. Educ.*, vol. 31, no. 4, pp. 512–528, 2021.
- [2] J. Burstein, J. Tetreault, and S. Andreyev, “Using automated essay scoring to assess writing skills,” in *Handbook of Automated Essay Evaluation*. New York, NY, USA: Routledge, 2013, pp. 55–67.
- [3] Google, “InterviewWarmup,” [Online]. Available: <https://grow.google/certificates/interview-warmup/>
- [4] S. Li and W. Deng, “Deep facial expression recognition: A survey,” *IEEE Trans. Affect. Comput.*, vol. 13, no. 3, pp. 1195–1215, Jul.–Sept. 2022.
- [5] J. Brusilovsky and E. Millán, “User models for adaptive hypermedia and adaptive educational systems,” in *The Adaptive Web*. Berlin, Germany: Springer, 2007, pp. 3–53.
- [6] J. K. Burgoon, L. K. Guerrero, and K. Floyd, *Nonverbal Communication*. New York, NY, USA: Routledge, 2016
- [7] H. Zhou and C. Leung, “Artificial intelligence in recruitment and selection: A review and research agenda,” *Int. J. Human Resource Manag.*, vol. 32, no. 20, pp. 4180–4209, 2021.
- [8] D. Worsley and R. Blikstein, “Multimodal learning analytics: Enabling the future of learning through multimodal data analysis,” in *Proc. ACM Int. Conf. Learning Analytics & Knowledge (LAK)*, New York, NY, USA, 2018, pp. 96–105.
- [9] T. Nguyen, T. Do, and Q. Tran, “Emotion-aware AI systems for education: A review,” *Comput. Educ.*, vol. 194, p. 104678, 2023.
- [10] S. Black and D. van Esch, “Artificial intelligence in hiring: Assessing impacts on fairness, ethics, and transparency,” *Bus. Ethics Q.*, vol. 32, no. 3, pp. 289–310, 2022.

